Moving Targets Detection for Video SAR Surveillance Using Multilevel Attention Network Based on Shallow Feature Module

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***Abstract*— In this article, a novel method for the moving target detection through multilevel spatial and channelwise attention network based on shallow feature channel module (MSCA-SFCM) is presented, and the circular spotlight (CSL) video synthetic aperture radar ground moving target indication (Video-SAR-GMTI) mode of the Nanjing University of Aeronautics and Astronautics miniature SAR (NUAA MiniSAR) system is introduced. However, due to the lack of moving target samples, MSCA-SFCM cannot be directly applied to the CSL Video-SAR-GMTI mode in the real system. To this end, this article proposes a training sample library construction scheme for moving targets of high verisimilitude. In this scheme, based on the radar system parameters, after the traversal of moving target parameters and SAR imaging, the scattering line characteristic of all possible moving targets under the current system parameters is simulated and then used for MSCA-SFCM network training. Afterward, the properly trained network can be used for moving target detection in real radar data. The effectiveness of the proposed method is verified by the NUAA MiniSAR system.**

***Index Terms*— Ground moving targets detection, multilevel spatial and channelwise attention network based on shallow feature channel module (MSCA-SFCM), training sample library construction, video synthetic aperture radar (video SAR).**

I. Introduction

IN THE future, given the accelerating pace of the war, the battlefield environment will be complex and changeable. On the battlefields of the future, the behaviors of multitudes of warfighters and adversaries have to be dynamically detected, identified, and characterized in real time. As a result, the military detection sensors have to evolve rapidly to meet the challenge posed by various highly mobile military targets, such as self-propelled artillery, vehicle-mounted missiles, armored vehicles, and mobile radar and tanks [1], [2], [3], [4], [5], [6], [7]. When fighting against enemy ground moving targets and providing fire support for ground troops, ground attack drones, helicopters, and other military platforms can employ photoelectric imaging sensors to detect and track the ground moving targets. However, the long-range detection performance of the photoelectric sensors depends highly on the light condition, which is usually far from satisfactory on the battlefield due to the consistent presence of smoke and dust. In contrast, radar has been widely acclaimed for its reasonably robust performance in highly adversarial environments and has played vital roles in many battles, e.g., the 78-day Kosovo war [8].

Traditional synthetic aperture radar (SAR) cannot meet the needs of real-time target monitoring in future battles, but the video SAR system can realize high frame rate and high-resolution ground moving target monitoring [9], [10], [11]. The video SAR system has the advantages of high imaging frame rate, high carrier frequency, and small antenna size, which can continuously observe ground moving targets all-day long and obtain important information, such as target motion parameters. Therefore, video SAR has attracted much attention in the field of moving target detection for its superior system performance.

In addition, many solutions [12], [13] have been proposed to improve the performance of target detection. After SAR imaging, due to the Doppler effect, the moving target component will shift in an azimuth direction and leave a shadow at its real position, and target motion produces an electromagnetic scattering line in the video screen. The bright line is the effect of defocusing of the moving target; hence, its position is different from the actual trajectory [14]. Several methods exploiting the characteristics of the shadow have been proposed to detect and track moving targets in the video SAR system [15], [16]. Besides, compared with the traditional SAR, the video SAR system is able to obtain dozens or even hundreds of sequential SAR images of the same scene, which can be exploited for moving target detection. Therefore, novel methods for background extraction have been proposed for sequential SAR images containing moving targets based on the mixed Gaussian background model [17]. By exploiting the characteristics of the moving target and the extracted background, the moving target component can be detected in the video SAR system [18], [19].

The robustness of the aforementioned traditional video SAR moving targets detection algorithm directly depends on the registration accuracy of SAR images between adjacent frames.

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Low registration accuracy will reduce the detection probability while increasing the false alarm probability. However, it is very difficult to apply high precision SAR image registration in the real world, especially for unmanned aerial platforms.

In recent years, some deep-learning-based algorithms exhibited reasonably robust performance against SAR image registration errors [20], [21], [22], [23], [24], [25]. Therefore, deep learning is employed in this work to solve the challenging technical problems in video SAR moving targets detection. In [21], a novel video SAR moving targets shadow detection method is proposed based on the convolutional neural network (CNN), in combination with the sliding window density clustering algorithm and the bidirectional long short-term memory network (Bi-LSTM). This new method has been verified by the video SAR data released by the Sandia National Laboratory.

A. Motivations

It should be noted that the shadow of moving targets in the video SAR system is an unstable characteristic in real-world applications. The shadow of the moving target of interest might deteriorate and fade into the background due to many factors, such as moving target speed, radar transmit power, noise figure, and background clutter power.

In this article, the circular spotlight (CSL) video SAR ground moving target indication (Video-SAR-GMTI) mode of the Nanjing University of Aeronautics and Astronautics miniature SAR (NUAA MiniSAR) system is introduced. When our NUAA MiniSAR system operates in the CSL Video-SAR-GMTI mode, the low transmit power of the system makes it difficult to discriminate the shadows of the moving targets from the background clutter due to the low contrast ratio. Therefore, it is very difficult and challenging for the NUAA MiniSAR system to detect moving targets by exploiting the shadow information when operating in the CSL Video-SAR-GMTI mode. However, in addition to the shadow feature, there is another important feature of moving target in the CSL Video-SAR-GMTI mode. That is, the effect of defocusing on the moving target will cause the bright line in the Video-SAR image, and the position of the bright line is different from its actual trajectory due to the Doppler effect. The bright line feature can be fully used to realize moving target detection in the CSL Video-SAR-GMTI mode of the NUAA MiniSAR system. Therefore, in the designed signal processing chain of the CSL Video-SAR-GMTI mode, the first step and one of the most important steps is the moving target detection by the bright line features based on a specially designed deep learning network; this is the motivation and also the key content of this article.

According to our plan, after initial moving target detection (single-frame detection results of CSL Video-SAR-GMTI), the time sequences of the Video-SAR images can be fully utilized to realize the following two applications. The first application of time sequences information is to reduce the number of false alarms and missed targets. The second application is to find the actual position of the detected moving targets based on the changed position of the detected bright line between adjacent frames.

B. Contributions

It is well known that, when applying deep learning methods for moving target detection in a real radar system, the first problem is that there are not enough moving targets samples for the data training, especially for the moving targets on the ground. With a small sample for neural network training, the detection performance and generalization ability will be greatly degraded. Therefore, the first contribution of this article is that we propose to construct the training datasets with the parameter traversal method. In the process of construction, all possible morphology and characteristics of moving target are included in the training datasets; hence, the trained neural network has good adaptability and generalization ability. In the process of simulated training datasets construction, we propose a step-by-step parameter traversal method and the concept of equivalent coefficients of time-varying distance slant, which can greatly reduce the workload of simulated training datasets construction. Experiment results show that the trained network with the simulated training datasets can well detect the actual moving targets in the CSL Video-SAR-GMTI mode, which demonstrates that the moving targets simulation approach can be one of the methods to alleviate the small sample problem when applying the deep learning method for target detection in radar signal processing.

Besides, for target detection in optical images, with the deepening of the network layer, the learning ability of target characteristics can be reinforced, and the detection accuracy can be improved to a certain extent. However, for CSL Video-SAR-GMTI datasets, the features of SAR images are relatively simple compared with optical images, and some features are not needed to be extracted in the network. Therefore, the research of the shallow layer network is considered to strengthen the performance of targets detection, and we propose a novel multilevel spatial and channelwise attention network based on shallow feature channel module (MSCA-SFCM) for moving target detection in the CSL Video-SAR-GMTI mode of our NUAA MiniSAR system. It is worth noting that the addition of spatial and channelwise networks allows the module to focus and learn the features and shapes of moving targets that we are concerned about. The effectiveness of the proposed network is verified by the real radar data collected by the self-developed NUAA MiniSAR system.

The third contribution of this article is that we briefly present the moving target relocation method utilizing inter-frame information of bright lines in inter-CSL Video-SAR-GMTI mode and give some initial moving target relocation results in one flight test to show the moving target relocation ability of the CSL Video-SAR-GMTI mode.

C. Article Organization

This article is organized as follows. Section II introduces the system parameters and the measurement results of the X-band video SAR system (Video-SAR-GMTI), which consists of the system parameters and the measured data results. The basic theoretical background of the proposed method framework is provided in Section III, which includes the radar platform model corresponding to the CSL model and the polar

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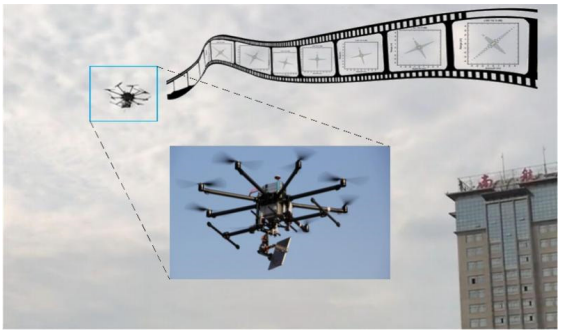


Fig. 1. UAV and the Video-SAR-GMTI system in the experimental scenario [27].

format algorithm (PFA). In addition, the method for reducing the number of ergodic parameters is presented. Section IV introduces a method for integrating the stationary background clutter and the moving targets in the echo domain, along with the step-by-step procedures for the construction of the simulated training dataset. Then, the detailed description of the proposed MSCA-SFCM network is presented in Section V. Section VI elaborates on the network training process, compares the moving targets detection results among different networks, and also introduces the initial scheme and results of moving target relocation based on time sequences information of the detected bright line feature.

II. Video-SAR-GMTI System

Compared to the conventional SAR, miniature SAR (MiniSAR) [26] features low power, lightweight, and small size and, hence, is more suitable for civil and public services, such as ground surveillance/reconnaissance and moving targets detection. The existing MiniSAR systems support operation at multiple frequency bands, which includes C-, X-, Ka-, and W-bands. In 2018, an X-band MiniSAR system was developed and tested by the research team at the College of Electronic and Information Engineering, NUAA [27].

A. CSL Video-SAR-GMTI Parameters

The X-band NUAA MiniSAR system consists of a phased-array antenna, a transceiver radiating linear frequency-modulated continuous waveforms, and a signal processing unit. It is capable of operating in a number of modes, which includes stripmap SAR, spotlight SAR, SAR-GMTI, and WAS-GMTI. Fig. 1 shows the unmanned aerial vehicle (UAV) and the Video-SAR-GMTI system in the experimental scenario. The main parameters are listed in Table I, which are used in SAR image sample simulation in Section IV.

B. CSL Video-SAR-GMTI Mode of NUAA MiniSAR System

In 2019, based on the NUAA MiniSAR system, the Video-SAR-GMTI mode was developed, and measured data were employed for testing, with the corresponding video SAR imaging results shown in Fig. 2. Fig. 2(a) shows the optical image of the radar radiation scene and presents the imaging results of the same scene from different angles. In the radar

TABLE I
Video-SAR-GMTI System Parameters

Parameter	Value	Unit
Carrier frequency	9.7	GHz
Rate of frequency modulation	4.5×10^4	GHz/s
Sampling frequency	0.54	GHz
Pulse repetition frequency	1000	Hz
Vehicle speed	6	m/s
Altitude of the carrier	289	m
Range resolution	0.33	m
Azimuth resolution	0.25	m
Frame rate	2	fps

imaging experiment, the partial imaging results of the moving car in Fig. 2(a) are shown as the arcs in the red box in Fig. 2(b)–(h).

Since the system operates in the X-band, the aperture multiplexing technology and circular SAR imaging mode are jointly exploited to generate the video SAR product. According to the principle of video SAR imaging, the moving target leaves a shadow feature in its real position, and the shadow is affected by the target's speed, the noise power, the background clutter, and other factors. When MiniSAR operates in the CSL Video-SAR-GMTI mode, the low transmit power of the system makes it difficult to discriminate the shadows of the moving targets from the background clutter due to the low contrast ratio between the two. Therefore, it is challenging for the NUAA MiniSAR system to detect moving targets by exploiting the shadow information when operating in the CSL Video-SAR-GMTI.

As is well known, in an SAR image, the moving target deviates from its real position, and the defocused scattering line produces a bright line as a result of range migration, the Doppler effect, and azimuth mismatch. Specifically, in our experimental mode, when the UAV platform moves in a circle, the radial velocity of the moving target relative to the radar system changes constantly so that the moving target generates a constantly changing bright line in the video SAR image. It can be seen in Fig. 2 that rich features are provided by the bright line characteristic for the video SAR, which can be extracted and exploited by deep neural networks for moving target detection.

III. Moving Target Simulation Based on Parameter Traversal Method for CSL Video-SAR-GMTI Mode

Given that the parameters of the SAR system are fixed, the shape of the moving target in the scene is determined by six parameters: the azimuth position, the range position, the azimuth velocity, the range velocity, the azimuth acceleration, and the range acceleration. To account for various detection scenarios, it is necessary to traverse these parameters to cover all the possible shapes of moving targets, which will lead to a heavy computational burden. Therefore, an effective approach

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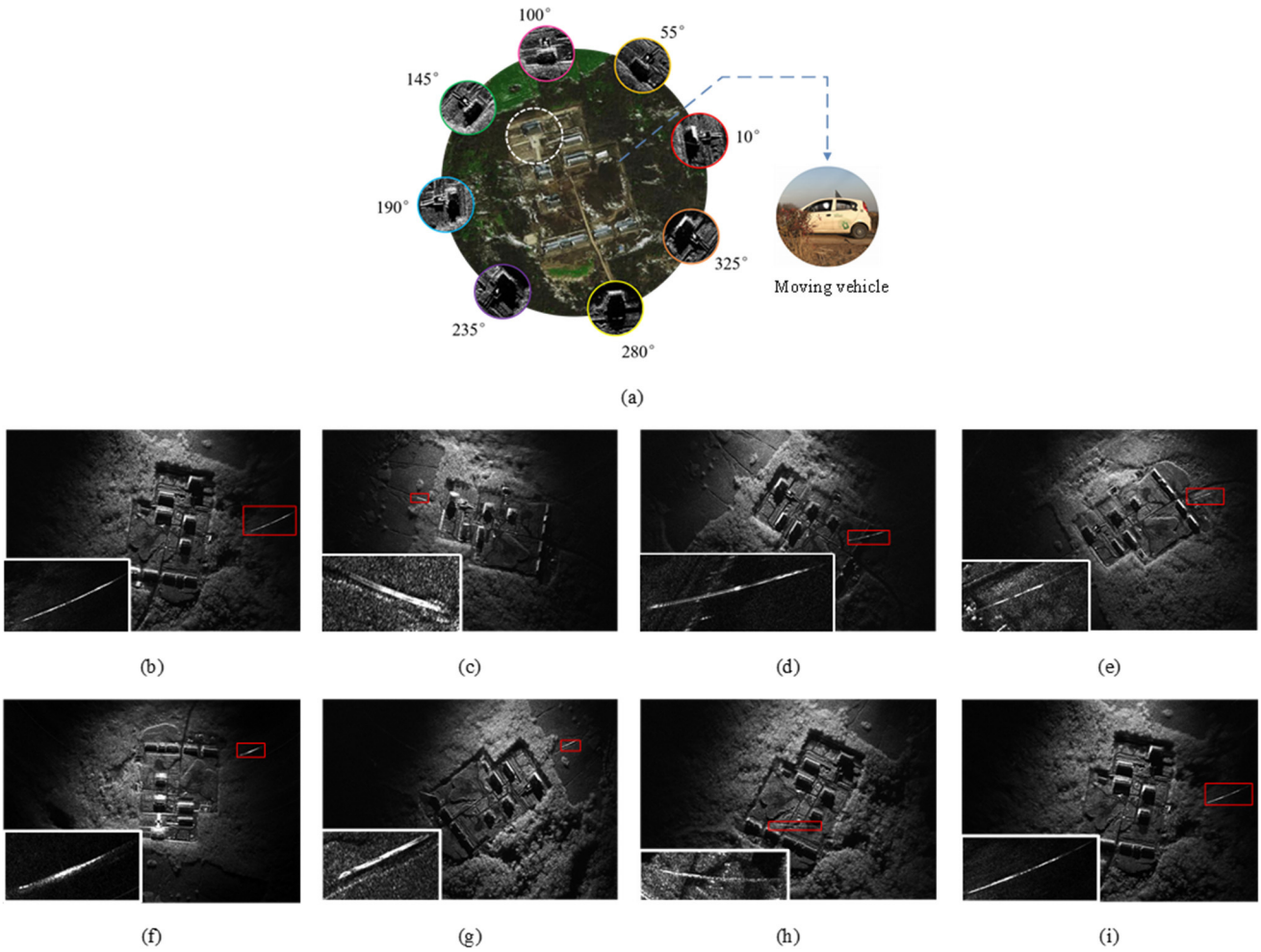


Fig. 2. Imaging results of CSL Video-SAR-GMTI. (a) Imaging results of target scenes from different angles [27]. (b)–(i) Imaging results of the scene of interest.

for the computational-load reduction in the process of training dataset construction is proposed in this section.

The special advantage of our proposed method in comparison with other simulation methods lies in that we propose a step-by-step parameter traversal method and the concept of equivalent coefficients of time-varying distance slant, which can greatly reduce the workload in the process of simulated training datasets construction. Besides, through the proposed parameter traversal method, all possible morphology and characteristics of moving targets are included in the constructed training datasets, and then, the trained neural network has good adaptability and generalization ability. As far as we know, the aforementioned method and concept have not been proposed in the field of Video-SAR moving target detection.

A. Echo Model of CSL Video-SAR-GMTI Mode

Deep neural networks have been widely applied in the field of video SAR moving targets detection and have exhibited outstanding performance. However, the performances of these networks are limited due to the lack of high-quality video SAR data that could be used for proper network training. Therefore, the network proposed in this work is trained with a high-fidelity experimental training dataset obtained via computer simulations and is tested with the measured video SAR data.

The linear frequency modulation continuous wave (LFMCW) transmitted by the SAR can be expressed as

$$S_0(\tau) = A\exp(j2\pi f_0\tau)\exp\left(j\pi k\tau^2\right) \tag{1}$$

where A represents the amplitude of the signal, τ represents the time index in fast time, f_0 is the carrier frequency, and k is the chirp rate. The received signal is a time-delayed version of the transmitted signal, which can be expressed as [28], [29]

$$r(\tau) = A'\exp(j2\pi f_0(\tau-\tau_i))\exp\left(j\pi k(\tau-\tau_i)^2\right) \tag{2}$$

where A' is the complex amplitude accounting for the propagation effects and τ_i is the time delay given by the moving target

$$\tau_i = 2R(t)/c \tag{3}$$

$R(t)$ represents the distance between the radar platform and the moving target, and c is the speed of light. With dechirp

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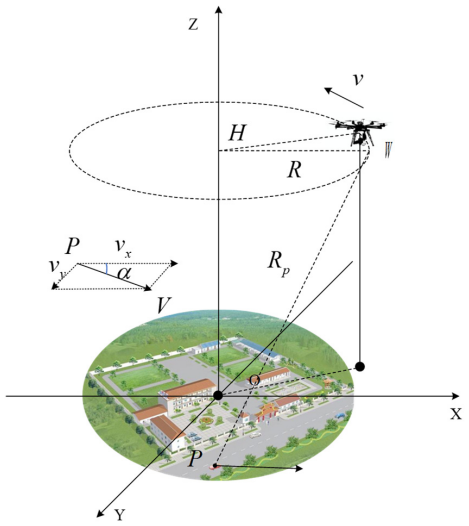


Fig. 3. Schematic of the geometric imaging model of CSL video SAR.

compression, the received signal is given by

$$\begin{aligned} S(\tau) &= r(\tau)S_{\text{ref}}^{*}(\tau) \\ &= \text{rect}\left(\frac{\tau-\tau_i}{T_p}\right)\exp(j2\pi f_0(\tau_0-\tau_i)) \\ &\quad\times\exp\left(j\pi k\left((\tau-\tau_i)^2-(\tau-\tau_0)^2\right)\right) \end{aligned} \tag{4}$$

where rect$(\cdot)$ is the rectangular window function and T_p is the pulsewidth of the radar transmitting signal.

$$S_{\text{ref}}(\tau) = \exp\left(j\pi k(\tau-\tau_0)^2\right)\exp(j2\pi f_0(\tau-\tau_0)) \tag{5}$$

where $S_{\text{ref}}(\cdot)$ represents the dechirp reference signal, $\tau_0 = 2R_p/c$, and R_p is the distance between the radar and the target. The beat signal is a single-frequency signal, and the signal frequency is proportional to the target distance. In this case, the Fourier transform is directly applied to the signal for pulse compression.

Inserting (3) into (4), the echo of the moving target can be calculated by

$$\begin{aligned} S(t,\tau) &= \text{rect}\left(\frac{\tau-2R(t)/c}{T_p}\right) \\ &\quad\times\exp\left(j\pi k\left(\left(\tau-2R(t)/c\right)^2-(\tau-\tau_0)^2\right)\right) \\ &\quad\times\exp\left(j2\pi f_0\left(\tau_0-2R(t)/c\right)\right). \end{aligned} \tag{6}$$

If the target is moving, the slant distance will change with t, which will affect the echo value. Therefore, the influence of the motion parameters on the slant distance is also considered in the traversal shape of the moving target in the scene.

Fig. 3 shows the schematic of the geometric imaging model of CSL video SAR. In Fig. 3, H, R, and v represent the altitude, radius, and velocity of the radar platform, respectively. The radar moves along the trajectory direction, and when it arrives at the aperture center, the target is at the position P, and the distance between the radar and the target is R_p. With the X-axis, the Y-axis, and the Z-axis representing the range, the azimuth, and the height dimension, respectively, the velocity of the moving target V can be decomposed into the range velocity v_x and the azimuth velocity v_y. The accelerations in the azimuth and range directions are set as a_x and a_y, respectively, and α in Fig. 3 is the angle between the moving target's velocity and the X-axis.

According to the geometric imaging model, the slant distance between the platform and the moving target can be expressed as

$$\begin{aligned} R(t) &= \sqrt{\left(R\cos\left(\frac{vt}{R}\right)-x_t\right)^2+\left(R\sin\left(\frac{vt}{R}\right)-y_t\right)^2+H^2} \\ x_t &= x+v_xt+\frac{1}{2}a_xt^2 \\ y_t &= y+v_yt+\frac{1}{2}a_yt^2 \end{aligned} \tag{7}$$

where x_t, y_t represents the position coordinates of the moving target at time t. When $t = 0$, the position coordinate is $(x, y, 0)$, and the slant distance is given by $R_p = ((R-x)^2+y^2+H^2)^{1/2}$. Therefore, with the approximations $\cos\theta \approx 1$ and $\sin\theta \approx \theta$, the slant distance is given by

$$R(t) = \sqrt{(R-x_t)^2+(vt-y_t)^2+H^2}. \tag{8}$$

After the moving target echo is obtained, PFA is implemented based on the detection scenario under concern. In this process, the geometric distortion is corrected by calibration at pixel level [30], [31], [32]. The detailed procedures of the PFA and the geometric distortion correction are shown in Fig. 4. Following the steps specified in Fig. 4, the corrected imaging results of targets can be obtained.

B. Moving Target Simulation Based on Parameter Traversal Method

Taking into consideration of (7) and (8), the shapes of the moving targets are affected by six variables, namely, the range and azimuth velocities v_x, v_y, accelerations a_x, a_y, and position coordinates x, y. To reduce the computational complexity, the Maclaurin expansion is used to reduce the number of variables that need to be traversed. Specifically, (7) is expanded into a Maclaurin series [33]. By keeping only the first- and second-order terms of t and discarding the higher order terms, the slant distance can be expressed as

$$R(t) \approx R_p + \text{A}_1t + \text{A}_2t^2 \tag{9}$$

where A_1 and A_2 are the equivalent coefficients of slant distance, which are given by

$$\begin{aligned} \text{A}_1 &= \frac{(v_y-v)y-v_x(R-x)}{R_p} \\ \text{A}_2 &= \frac{(v_y-v)^2+a_yy-a_x(R-x)+v_x^2}{2R_p}-\frac{\text{A}_1^2}{2R_p}. \end{aligned} \tag{10}$$

When simulating the moving target morphology with a fixed set of radar system parameters, the values of the parameters H, R, and v of the radar platform should be fixed. The simulation of the target echoes should include all the possible shapes of the moving targets by traversing the parameters A_1 and A_2. It was found from the preliminary experimental results that the

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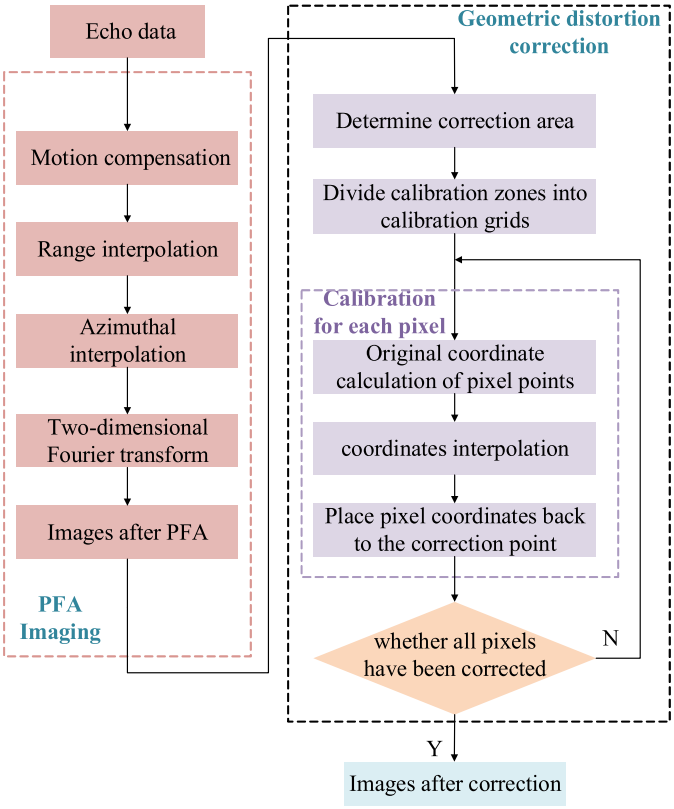


Fig. 4. Flowchart of signal processing of imaging.

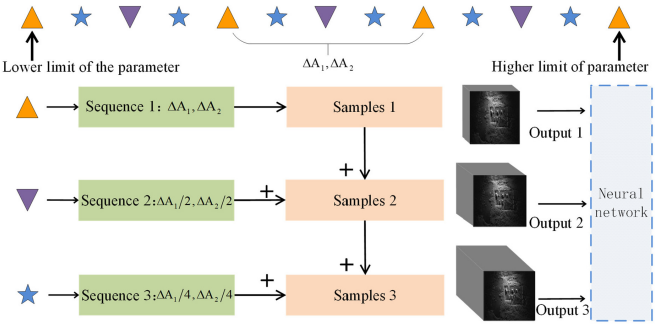


Fig. 5. Equidistant step-by-step traversal of parameters.

computational load increases with the decrease of the parameter step intervals. Moreover, although the detection probability becomes higher with reduced step intervals, the false alarm rate goes up as a tradeoff. Therefore, both the computational efficiency and the moving target detection accuracy should be considered when selecting the optimum step interval of parameters for practical application.

In this article, the step-by-step traversal method with discretized equally spaced parameters is adopted to synthesize the echoes of the targets of interest, as shown in Fig. 5. In the following, the process of sampling interval selection in three successive steps is offered as an example to explain the procedure of sample library construction. The upward-pointing triangle, the downward-pointing triangle, and the pentagram correspond to the discretized values of A_1 and A_2 in three successive steps, and the sampling intervals are within the range of the lower and higher limits of the parameter. According to the sampling relation in Fig. 5, the upward-pointing triangle corresponds to the interval of $\Delta A_1, \Delta A_2$ forming the dataset of sample 1. A new sample library is obtained by selecting the new parameters (the downward-pointing triangle) at the mid-points of two adjacent samples with intervals of $\Delta A_1/2, \Delta A_2/2$. Combined with the dataset of sample 1, the dataset of sample 2 is formed. Afterward, data samples with intervals of $\Delta A_1/4, \Delta A_2/4$ can be obtained in a similar manner.

The performance of the deep neural network depends highly on the amount and the quality of the labeled samples that could be used for network training. Generally speaking, the deep neural network exhibits higher detection probability when trained with large amounts of high-quality samples covering all kinds of potential targets in various detection scenarios. Meanwhile, the construction of an overextensive training dataset could lead to heavy computational burdens. Moreover, it is also found through experiments that the increase in sample sets will increase the false alarms to some extent and seriously affect the efficiency of the proposed approach. Therefore, the optimal size of the simulation sample set is determined by continuously increasing the sample set for training and detection with the step interval method presented above so that the detection probability is maximized, while the computational burden and the false alarm rates are kept in control.

The values selection of A_1, A_2 in our experiment is introduced in Section VI.

IV. Construction of High-Fidelity Moving Training Dataset

To accomplish a typical ground surveillance/reconnaissance mission, the target detection algorithm must be capable of distinguishing the targets of interest from all kinds of background clutter. When the background clutter is strong, part of the moving target trajectory might be buried in the background, which leads to false alarms and missed detections.

Therefore, in this section, we propose an effective method for high-fidelity SAR training dataset construction with the impacts of clutter taken into account.

A. High-Fidelity Training Dataset Construction Method

The flowchart summarizing the key steps of training dataset construction is shown in Fig. 6.

To begin, several SAR imaging scenes are randomly selected considering the NUAA MiniSAR system parameters, and then, the background should be denoised, rotated, and enhanced, aiming at adapting the training set to the complex background and simple background and other diverse situations. After that, the remaining background clutter is adaptively fused with the synthesized target echoes to account for various signal-to-clutter ratios. To enhance the flexibility and robustness of the deep neural network-based algorithm in various detection scenarios, an extensive training dataset containing all types of background clutter and targets has to be constructed. Therefore, we propose an effective method in the following for the seamless fusion of the synthesized

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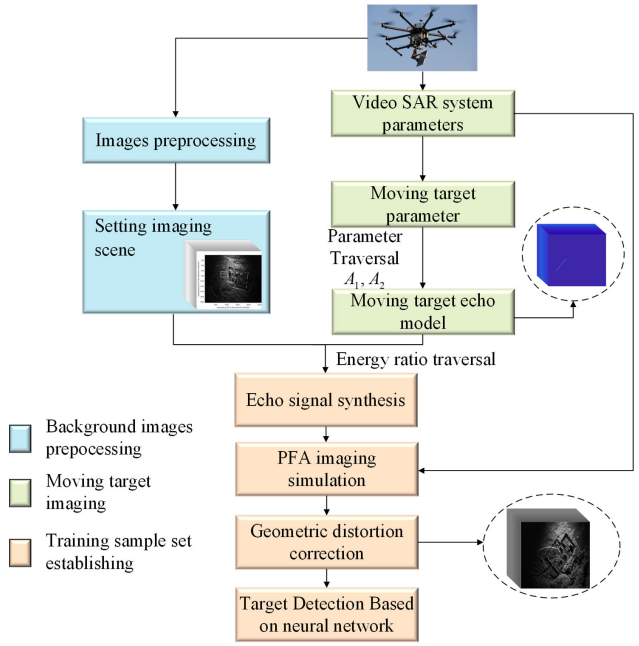


Fig. 6. Key steps of establishing a training sample set.

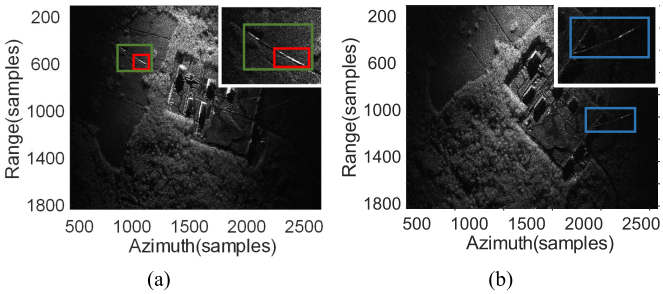


Fig. 7. Two examples of the synthesized video SAR images. (a) False alarms. (b) Missed targets. (The ground truth, the false alarms, and the missed targets are indicated by the green, red, and blue rectangles, respectively.)

background clutter and the moving target echoes, which is summarized in Table II.

We construct matrices to store the preprocessed video SAR background. Since the images obtained from the measured data are gray-scale images, the pixel values are between 0 and 255. Before moving on to the detailed image fusion algorithm, we define a new parameter, ER, as the energy of the echoes from the targets to that of the echoes from the background clutter. When the value of ER is low, part of the bright line in the video SAR images generated by the moving target might be concealed in the strong background clutter, leading to a discontinuous line in the SAR image. Therefore, if the effects of ER are not properly taken into account in the process of SAR image sample synthesis, the broken bright line generated by one target might be misjudged as two targets. To better illustrate this point, a pair of synthesized video SAR images are presented in Fig. 7, where the ground truth and the false alarm caused by the broken target are marked by green and red rectangles, respectively, while the missed targets are indicated with blue rectangles in Fig. 7(b).

To improve the target detection performance in the scenarios illustrated in Fig. 7, we expand the training data to account for different values of ER. The echo signals from the background clutter and the moving targets, which are represented by S_b and S_0, respectively, are synthesized separately. After that, the echo signals from the whole scene are obtained by adaptively combining these two parts according to the ER value of the target detection scene under concern. With the effects of clutter on the moving target accounted for, the synthesized SAR data samples are of high fidelity and match perfectly with those that would have been obtained in real-life measurement campaigns.

The simulated echo signals containing both the targets and the background clutter are then processed with matched filtering, motion compensation, interpolation, and geometric distortion compensation, after which an extensive set of training samples of high fidelity is obtained. It should be pointed out that the Kolmogorov–Smirnov (KS) [34] two-sample test is performed, and the test results show that the simulated data based on the aforementioned method and the actual testing data from the CSL Video-SAR-GMTI mode of our NUAA MiniSAR system belong to the same distribution. Due to space limitations, these results are not presented here.

B. Qualitative Analysis of High-Fidelity Training Dataset Construction Method

It is true that the simulated samples will be more in accordance with the real ones if we consider the radar backscattering characteristics of the extended moving targets. Many key factors influence the value of radar backscattering characteristics, including radar observation angles, background environments, the size of the moving targets, and so on. The simulation work will be rather complex to perform in real-world applications if we consider the radar backscattering characteristics of the extended moving targets since all possibilities should be covered to guarantee the generalization ability of the trained neural network. Besides, in this article, the proposed deep learning method is used to detect moving targets not to realize target recognition; some simplifications are made to construct the training datasets. The core idea of the simplification is that the shape feature plays more important roles in moving targets detection based on the deep learning method in the CSL Video-SAR-GMTI mode. At the same time, the shape feature of the moving target is mainly depending on the target's motion parameters and position parameters.

Therefore, the training datasets in this article are constructed with the parameter traversal method, which concentrates on motion parameters and the position parameters of the moving target. Then, the influences of observation angles, background environments, and moving targets' sizes on the proposed parameter traversal method are analyzed as follows.

The influence of observation angles mainly lies in two aspects. The first influence is that the main moving parameters (such as velocity and acceleration in range and azimuth directions) of the same moving target will change greatly in different observation angles due to the changes in relative position between the platform and the moving target. Therefore, the same moving target may appear with different morphological

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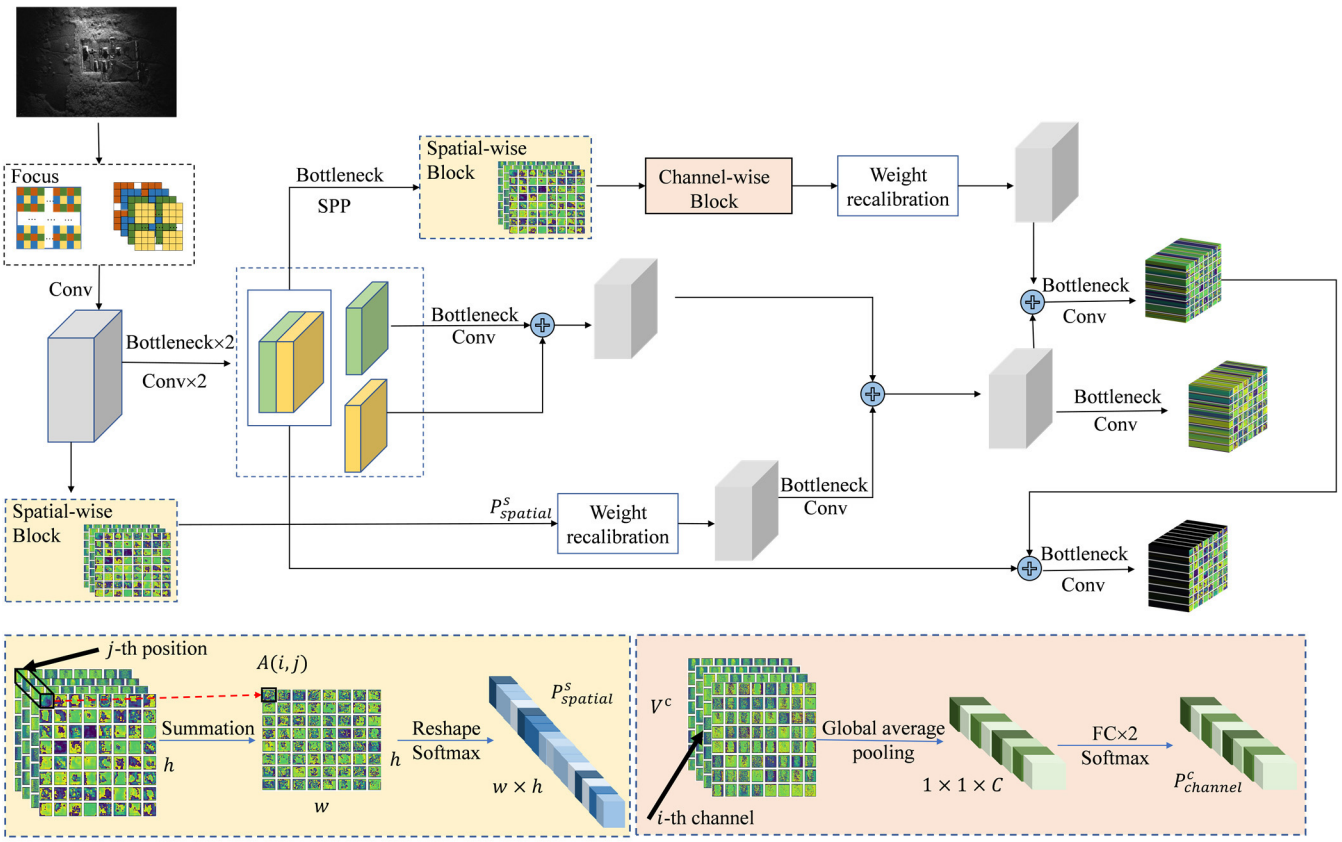


Fig. 8. Network structure of the MSCA-SFCM.

features, which can increase the detection ability of the moving target. The second influence of observation angles is that moving targets may have different radar cross sections (RCSs) from different observation angles. This may affect the signal-to-clutter ratio (SCR) together with the different background environments.

When the background environment is relatively complex, part of the "bright line" of the moving target after SAR imaging will be submerged in the background clutter, resulting in the discontinuity of the final imaging result. The influence of the discontinuity of the "bright line" and the corresponding solution has been described in detail in Section IV-A.

Finally, we analyze the influence of different moving targets' sizes. In the CSL Video-SAR-GMTI mode, moving targets are supposed as rigid construction; therefore, the simulation of the extended moving target can be equivalent to the summation of the simulated multiple points echo of the target. From the point of SAR imaging for the CSL Video-SAR-GMTI mode, if the moving targets have the same motion parameters but with different sizes, their shapes are mostly alike except slight thickness differences. Besides, the shape feature plays more important roles than the thickness feature in the processing of moving targets detection based on the deep learning method.

V. Target Detection Using MSCA-SFCM

Moving targets detection in remote sensing imagery with deep neural networks has been intensively studied in the literature. The algorithms for targets detection in remote sensing imagery could be loosely categorized into two types. The group of two-stage algorithms forms the first category, which includes CNN [35], Faster-RCNN [36], Mask-RCNN [37], and so on. These algorithms first generate candidate object-bounding boxes, based on which classification and regression are carried out sequentially. The second category consists of one-stage algorithms, such as You Only Look Once (Yolo) [38] and the single-shot multibox detector (SSD) [39]. Compared with two-stage algorithms, the one-stage algorithms demand less training time and could accomplish the detection/classification task in a more time-efficient manner. In this section, MSCA-SFCM is proposed for moving targets detection in video SAR.

Generally speaking, the shallow layers of the artificial neural networks (i.e., those closer to the input layer) function well in extracting low-level features (e.g., texture and edge) from the image samples, while the deeper layers are more capable of extracting high-level semantic information. However, overexcessively increasing the depth of the neural network would not necessarily lead to better detection performance, especially in the case of moving targets detection in SAR images, where the information content per pixel is much less than the corresponding one of the rich-colored optical imagery.

Shallow layer features are fed into the spatialwise attention module to obtain the weight information of different positions in the image, which is used for weight update in the next network layer. In the middle layer, the channelwise operation

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is carried out to update the weights for better detection performance.

A. Basic Structure of MSCA-SFCM

The MSCA-SFCM network structure model is shown in Fig. 8.

In the training process, in order to detect targets of different sizes, three prediction boxes of different sizes will be implemented for each input image. The whole network model is mainly composed of four parts.

1) MSCA-SFCM mainly adopts the bottleneck network structure to extract the deep features of the input images. Two 1×1 filters are employed in the bottleneck network to reduce and increase the feature dimension, respectively, thus reducing the number of parameters that are required to validate the robustness of computation reduction in the training process.
2) Perform the spatialwise operation on the shallow feature map to take sufficient advantage of the rich location and edges information in the shallow layers to obtain the weights in the feature map.
3) Partial cross-stage operation is introduced into the middle layer to appropriately mitigate the gradient descent problems caused by network depth and to get better access to the upper layer features.
4) The weights are continuously updated during the network training process by combining spatial and channelwise attention so that high-level critical discriminative features are extracted.

The mosaic data enhancement method and the focus module, which were originally proposed the YOLO networks, are adopted at the input end of the proposed network for data preprocessing. Mosaic data enhancement is an effective data augmentation technique by randomly scaling, clipping, and arranging the input images, which can effectively expand the data set. The key function of the focus module is to slice the input images for downsampling and channel expansion while reserving the complete image information. Consequently, it can effectively reduce the number of parameters, realize dimensionality reduction, and broaden the local reception field.

B. Multilevel Spatial and Channelwise Attention Network

When training the input image data, the network layers need to concentrate on the critical features by observing the calculated weights in each channel and each pixel in the feature map. Therefore, the multilayer spatial and channelwise attention module is proposed in this part for network training. Spatial attention and channelwise attention are combined in a flexible way to extract the low- and high-level features from the targets of interest at various depths of the neural network. MSCA-SFCM introduces the attention module in the following two stages of network training.

1) The feature map obtained from the shallow network is rich in edge and location information. For spatial attention operation, more concrete methods to obtain weight are shown in Fig. 8. Assuming that the size of the feature map in the network is $w \times h \times c$, the values of each pixel point in the multichannel feature maps are summed up to obtain a 2-D matrix with a size of $w \times h$. $A(i, j)$ in Fig. 8 denotes the summation of the activations along each channel. By reshaping the matrix as a column, the weight of each pixel is calculated as

$$\begin{aligned} a &= \tanh((W_s V + b_s) \oplus W_{\text{hs}} h_{t-1}) \\ P^s_{\text{spatial}} &= \text{soft}\max(W_i a + b_i) \end{aligned} \tag{11}$$

where W_s, W_{hs}, and W_i represent transformation matrices, which are used to map hidden states and image visual features to the same dimension. V denotes the input vector of dimension c. $\oplus$ represents the addition of matrix and vector. b_s and b_i are model biases. α is the weight of each pixel in the feature map.

With spatial attention applied, greater weights are assigned to the key parts in the primary feature map; hence, the neural network model can focus on the critical features of the target of interest.

2) Carry out spatial and channelwise attention operations for feature maps in middle layers. Note that the successful generation of high-level feature maps depends on the quality of feature maps at the low level, i.e., it is only possible for the deeper layers to extract the target's high-level features once the edge information for the region of interest is successfully extracted by the shallow layers. Therefore, spatial attention and channelwise attention are combined in the middle layers.

The input feature map vector V contains the weight coefficients at different positions through spatial attention. Assume that the feature map vector after weight updating is V^c, which is fed into channelwise attention. The module is primarily composed of function blocks for squeezing, compressing, and reweighting. The feature maps in the input vectors are compressed with global average pooling so that each feature map is represented by a specific value, which to some extent corresponds to the global receptive field of the feature channel. Each value in the resulting $1 \times 1 \times c$ vector is given by

$$d_c = \frac{1}{w \times h} \sum_{i=1}^{w} \sum_{j=1}^{h} a_c(i, j) \tag{12}$$

where $a_c(i, j)$ represents the value of each position on the feature map and d_c is the eigenvector after the operation. The weight of each channel can be represented as

$$\begin{aligned} b &= \tanh\left(\left(W_c \otimes V^c + b_c\right) \oplus W_{hc} h_{t-1}\right) \\ P^c_{\text{channel}} &= \text{soft}\max\left(W_i' b + b_i'\right) \end{aligned} \tag{13}$$

where W_c, W_{hc}, and W_i' represent the transformation matrices, b_c and b_i' are biases, and $\otimes$ denotes the outer product of vectors.

C. Shallow Feature Channel Module

In the process of network training, feature maps corresponding to different depths contain different information. Shallow feature maps mainly contain low-level information (such as edges, colors, and primary positions of targets). The deep feature maps contain advanced semantic information for

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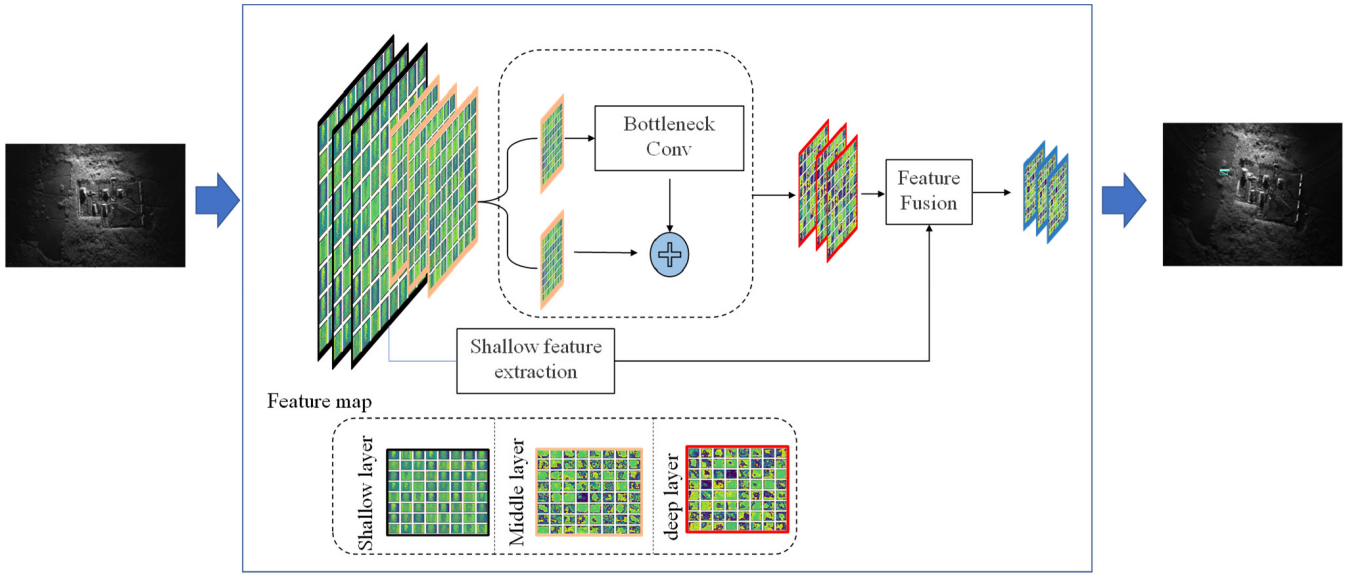


Fig. 9. Network structure of the SFCM.

targets detection, which is critical to targets recognition and classification. It is also worth noting that the pixels information of the target is relatively small in the video SAR moving targets detection system, which has a high requirement for the details of images. Furthermore, the accurate location of the target is also required for moving targets detection and classification in the SAR images. To ensure the generalizability of the proposed algorithm, we propose a novel neural network to integrate the shallow features with the high-level features obtained at the deeper layers. The shallow network model is shown in Fig. 9. The key steps for feature extractions in shallow layers are given as follows.

1) The weight value of each pixel is calculated after spatial attention in the shallow layer to update the weight. Then, concat with the feature map obtained from the deeper network for subsequent processing.

2) In the middle layers, partial cross-stage operation is carried out, and the input feature maps are divided into two parts according to a certain proportion. One part is passed on to the next level through the network structure chosen, and the other part goes through a shortcut. The partial cross-stage operation can partially retain the upper layer features, combine the information with the other part after the convolution operation, and, hence, can effectively enhance the learning ability of the neural network. In addition, the shortcut can effectively reduce the computational load and improve computational efficiency.

VI. Experiment

The proposed MSCA-SFCM is first trained by the simulated extensive SAR image dataset and then used for targets detection in the measured data obtained by NUAA MiniSAR. The performance of the proposed MSCA-SFCM is compared with that of the HOG + SVM, Faster-RCNN, Yolov5, and the MSCA, and a detailed performance analysis is carried out.

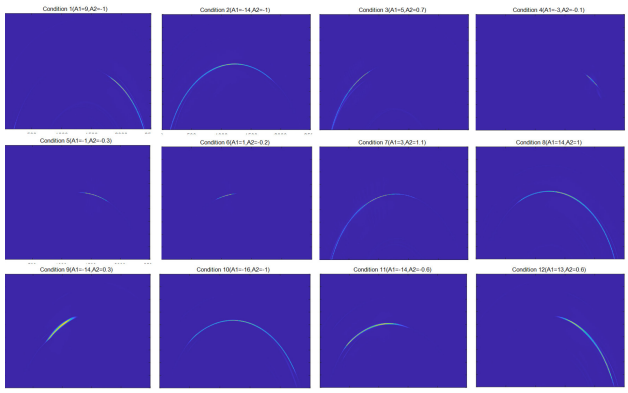


Fig. 10. Moving target energy in a single frame with different motion parameters.

A. Result of Simulated Video SAR Data

The simulated moving targets model is set as a 2-D rectangle in accordance with the actual scene. The echo signal of the moving targets is calculated through the summation of the echoes of all the scatters on the target model. Moving targets with different motion parameters are used to verify the robustness of the simulation.

In Section III, an approach of step-by-step parameter traversal has been discussed to simulate moving targets in real situations. Based on the moving parameters of most ground moving targets, the velocity is set between 0 and 80 km/h, while the acceleration is set between 0 and 2 m/s^2 in the process of training dataset construction. In this article, the deep learning method is adopted for moving targets detection mainly on the basis of the morphological characteristics of targets in SAR images. In order to reduce the workload of parameter traversal, only A_1 and A_2 are used to traverse the moving parameters in this article. Partial results of moving target energy in a single frame with different motion parameters are shown in Fig. 10.

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TABLE II
BACKGROUND FUSION ALGORITHM

Algorithm: Background Fusion

Input: Moving target echo signal + Background image after preprocessing
Output: Training dataset
Steps: The pixel value of the background clutter in the SAR images is $P(i,j)$

1. Construct a matrix to store the background clutter data and determine the scene size for radar imaging.
2. Normalize the background image pixel values as $P_x(i,j), P_x(i,j) = P(i,j)/255$
3. The normalized pixel values are taken as static target points to obtain the background clutter signals S_b with high fidelity.
4. Synthesize the moving target echo signal S_0.
5. Combine the background clutter and the moving target echo as $S = S_0 + \mathrm{ER} \times P_x \times S_b$.
6. Synthesize the SAR images with the PFA.

TABLE III
VARIATION RANGE OF PARAMETERS IN DIFFERENT TRAINING STAGES

	First stage	Second stage	Third stage
A_1	A_1^1 = [-16,-12,-8,-4, 0,4,8,12,16]	$A_1^2 = A_1^1$ U [-14,-10, -6, -2, 2, 6, 10, 14]	$A_1^3 = A_1^2$ U [-15,-13,-11, -9, -7, -5, -3, -1, 1, 3, 5, 7, 9, 11,13,15]
A_2	A_2^1 = [-0.9, -0.5,-0.1,0,0.1,0.5,0.9]	$A_2^2 = A_2^1$ U [-0.7, -0.3,0.3,0.7,1.1]	$A_2^3 = A_2^2$ U [-0.8, -0.6, -0.4, -0.2, 0.2, 0.4,0.6, 0.8, 1.0]
Samples	600	2000	6210

Note that, since the NUAA MiniSAR has limited operational height and speed, as a result, the coordinates and speed of moving targets have a great impact on the imaging results. Therefore, when synthesizing the training dataset corresponding to various target motion states, the shapes of scattering lines generated by the moving targets are complex and diverse, which makes it more difficult to detect targets from the measurement data obtained by the MiniSAR system than from those obtained by the traditional video SAR system.

B. Test Results of Simulated Video SAR Data

Table III lists the changing process of parameters during the step-by-step parameter traversal process. After the construction of the training dataset, there are 62, 200, and 621 moving targets with different shapes in the first, second, and third stages, respectively. When the moving target and the background clutter are fused in the echo domain, five energy ratios are traversed, and the values are selected as 0.5, 0.75, 1,1.25, and 1.5, and randomly set the target at any two positions in the scene. Since the Mosaic enhancement is conducted to MSCA-SFCM before the image is input into the network, the original image input in this article is no longer processed with random rotation and translation. Consequently, there are 620, 2000, and 6210 cooperative targets in the training sample databases of the three comparative experiments.

TABLE IV
DETECTION PERFORMANCE OF REAL TARGETS BASED ON DIFFERENT SIMULATION TRAINING SAMPLE SETS

Stage of training	Correctly detected targets	False alarms	Missed targets
First stage	158	135	469
Second stage	372	136	255
Third stage	534	95	93

During the network training process, a verification set will be run after each epoch to verify the training effect, so as to timely spot the problems with the model or parameters and make instant adjustments for the too slow growth rate of the parameter model mean average precision (mAP) or the loss function explosion. Moreover, the aforementioned application of the verification set can verify the general ability of the model. To prevent the overfitting of the training model, it is of vital significance to conduct a comprehensive comparison of the effects of the verification set and the training set. Therefore, the existing training samples should be divided. In this experiment, samples are divided into the training set and the verification set in a ratio of 8:2. Then, the training results were applied to the moving targets detection of the video SAR measured data collected by the CSL Video-SAR-GMTI mode of the NUAA MiniSAR system. After the frame division of the MiniSAR measured data, 514 effective images were fed to the network, and 627 effective targets were required for detection. Table IV shows the detection results of the measured data based on different video SAR simulation training sample sets with the MSCA-SFCM method.

It can be seen from Table IV that, with the decreased step length of parameter traversal, there are more diverse types of imaging forms of the simulated targets in the training sample set, conforming to the characteristics of the varied velocity and acceleration of targets, so this is closer to the sample set under actual conditions. The results of moving targets detection at the three stages are exhibited in Fig. 11, where (a), (c), and (e) use the same image for detection, and the detected images in (b), (d), and (f) are the same. The rectangular box marked with the confidence coefficient and label is the result obtained directly from the detection end of MSCA-SFCM. The red oval boxes marked outside the confidence indicate the correctly detected target, while the blue oval boxes highlight the missed targets, and the yellow oval box indicates the false alarm target.

According to the final detection results, the number of correctly detected targets increases gradually. It is proven that

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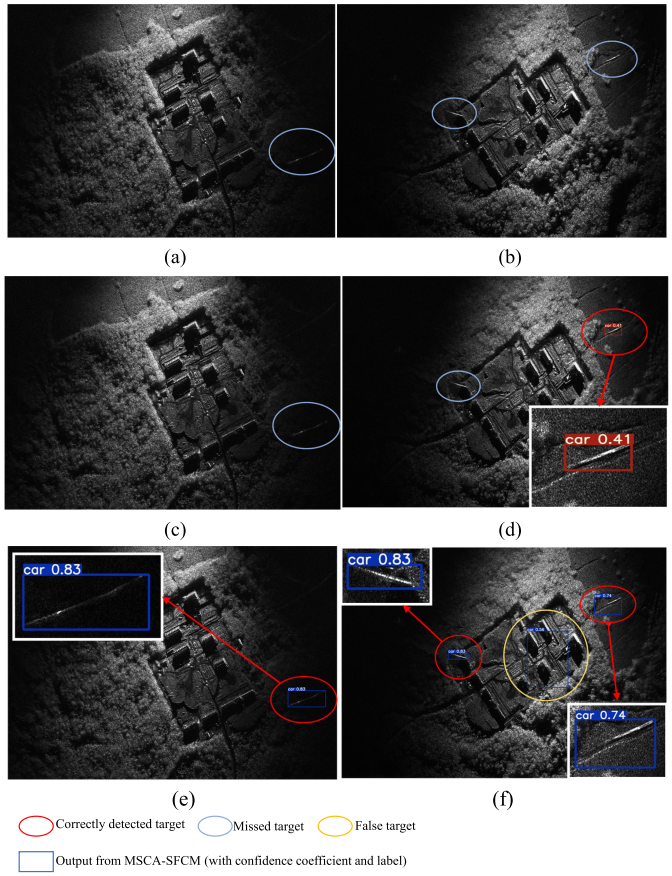


Fig. 11. Partial results at different detection stages. (a) and (b) Detection results in the first stage. (c) and (d) are in the second stage. (e) and (f) are in the third stage.

by increasing the simulation samples appropriately to cover the comprehensiveness of target shapes, the probability of moving targets detection can be effectively improved. There are also some false alarms that remain. The background clutter is the imaging of the stationary target, and some of their shapes are similar to the simulation imaging results of the targets in the imaging scene, thus resulting in false alarms. In addition, due to the different surface materials of the targets in the actual situation, there is an energy imbalance in the single frame of the target in the actual imaging results, which can be verified by the measured data of MiniSAR. In the process of training, false alarms can also be a trouble, but this problem can be eased by LSTM [40] and the clustering algorithm.

C. Detection Result With the Influence of ER

In video SAR images, background clutter in the scene may deteriorate the shapes of the scattering line, leading to a misjudgment of the targets. To prove the influence of the introduced parameter ER on the accuracy of final detection results, some experiments are conducted in this subsection. 621 target forms from the third stage mentioned in Section VI-B, are adopted and parameters ER were selected as [1], [0.5, 1, 1.5], [0.5, 0.75, 1, 1.25, 1.75], respectively, for target and background fusion. The three groups of simulated data were constructed as training sample sets of neural networks. Choose the MSCA-SFCM network as the training module; the experimental results obtained are shown in Fig. 12.

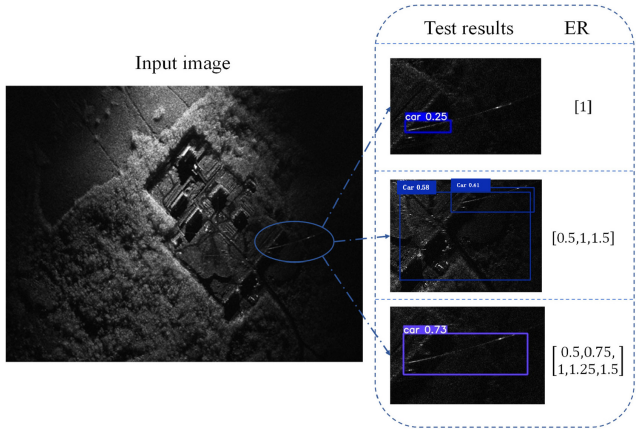


Fig. 12. Experimental results with different selected ERs.

TABLE V
COMPARISONS OF DETECTION PERFORMANCE ON THE MEASURED VIDEO SAR DATA

Method	Correctly detected targets	False alarms	Missed targets
HOG+SVM	312	284	315
Faster-RCNN	463	297	164
Yolov5	458	235	169
MSCA	502	165	125
MSCA-SFCM	534	95	93

The moving target in the image has discontinuity caused by the surrounding clutter. If ER is not introduced, only the partial scattering line can be detected. Set ER to 0.5, 1, and 1.5, and the scattering line can be completely framed, but there is still interference from other surrounding areas to the result, and it shows the overlapping phenomenon of multiple targets. By adding 0.75 and 1.25 to the range, the actual moving target scatterers can be well identified.

D. Test Results of Comparative Experiment

After constructing the high-fidelity training sample set, this article proposes MSCA-SFCM for the detection of moving targets. In order to reflect the performance of the proposed network and the traditional networks on moving targets detection, quantitative comparisons in terms of HOG + SVM, Faster-RCNN, Yolov5, MSCA, and MSCA-SFCM are performed based on the simulated training dataset, and the experimental results are compared and analyzed. All the simulation data at the third stage (in part B) are taken as the training set by the five network models to detect 627 targets in total. Statistical results of the aforementioned networks are shown in Table V.

Feeding each video SAR image into the trained HOG + SVM, Faster-RCNN, and Yolov5 network yields the preliminary detections, which contains 284, 297, and 235 false alarms and 315, 164, and 169 missed targets, respectively.

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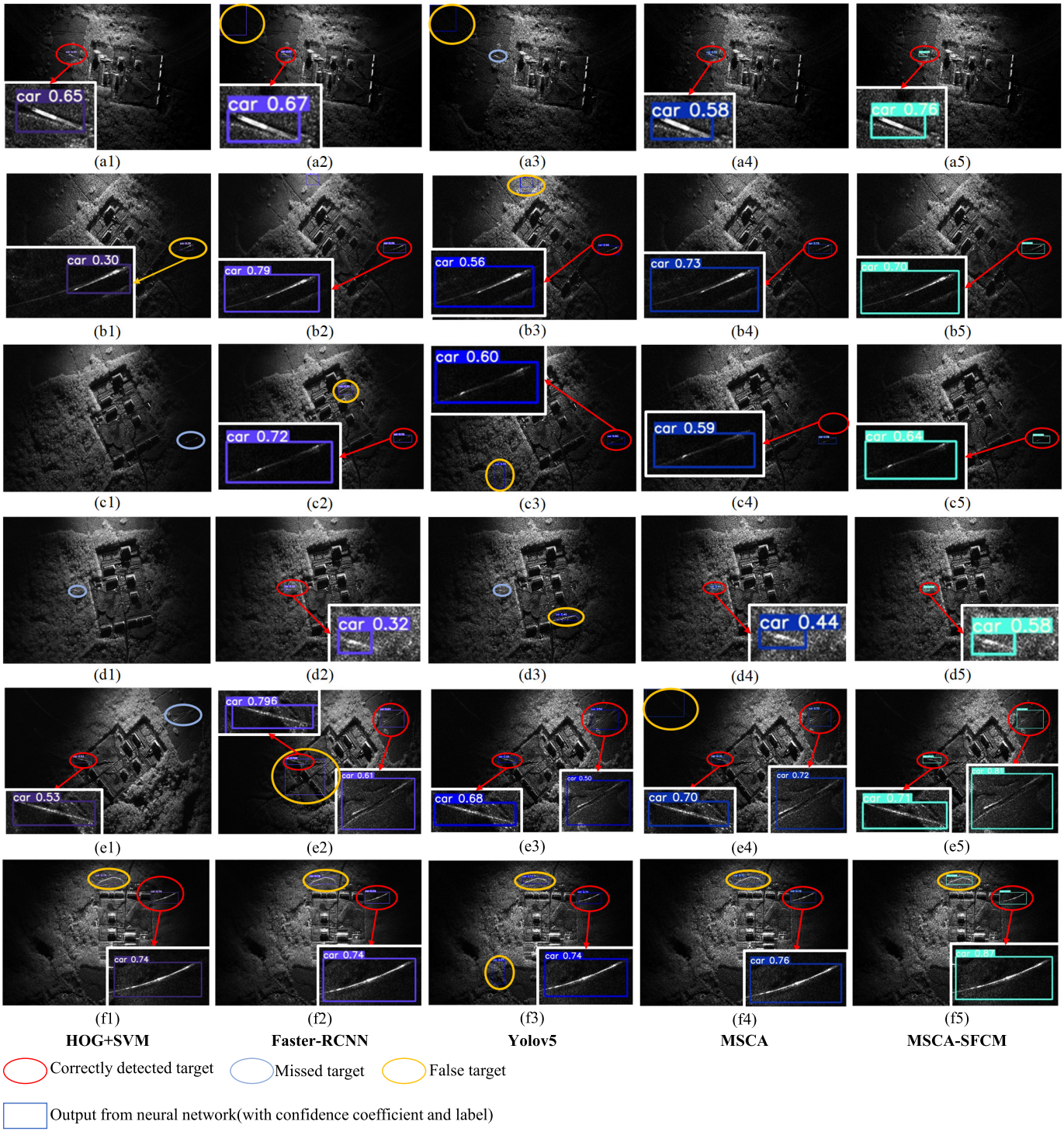


Fig. 13. Partial detection results of different networks. (Different rows indicate different SAR images of the CSL Video-SAR-GMTI mode.) (a1)–(f1) Output from HOG+SVM detector for frame 54, 143, 208, 254, 445, and 815, respectively. (a2)–(f2) Output from Faster-RCNN detector for frame 54, 143, 208, 254, 445, and 815, respectively. (a3)–(f3) Output from Yolov5 detector for frame 54, 143, 208, 254, 445, and 815, respectively. (a4)–(f4) Output from MSCA detector for frame 54, 143, 208, 254, 445, and 815, respectively. (a5)–(f5) Output from MSCA-SFCM detector for frame 54, 143, 208, 254, 445, and 815, respectively.

The HOG + SVM method has been widely used in targets detection, which utilizes the histogram of gradient or edge direction of each pixel in the image, finally forms a feature descriptor, and then uses SVM to judge the target. The core idea of HOG is that the shape of the detected local target can be described by the distribution of gradient or edge direction, which can better capture the local shape information and has good invariance to geometric and optical changes. However, due to the characteristics of gradient, HOG is relatively sensitive to noise and will be strongly affected by noise, resulting in a low correct detection rate of the targets. Faster-RCNN is a classical second-order algorithm, which has achieved good results in targets detection and targets classification in various fields. By using the feature extraction method, the

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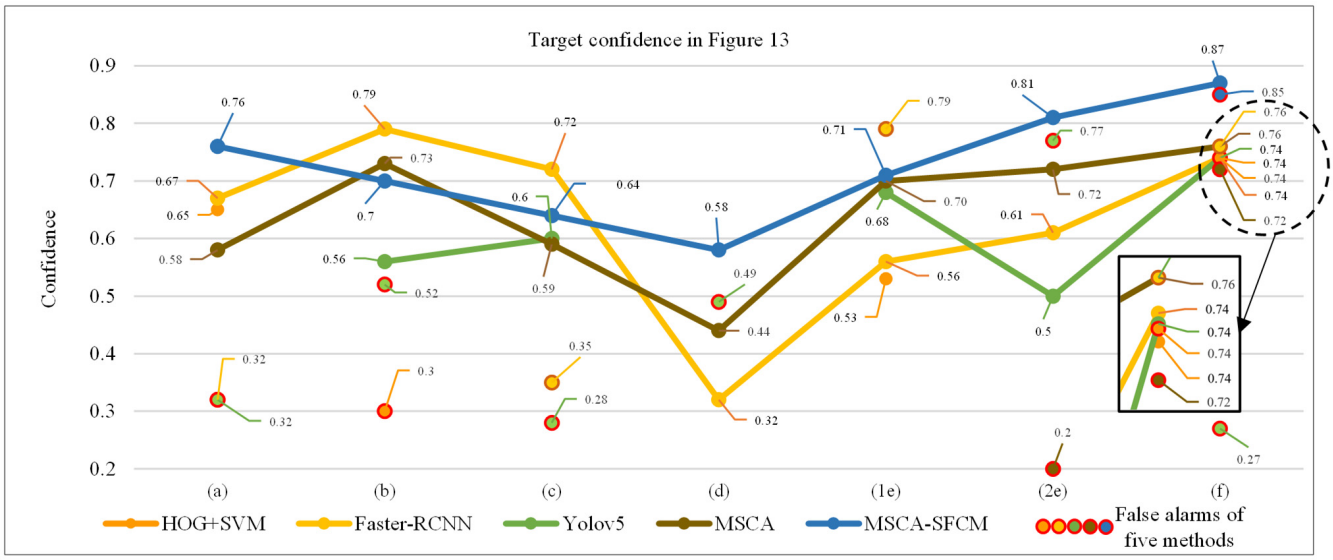


Fig. 14. Target confidence in different networks. (a) Target in frame 54. (b) Target in frame 143. (c) Target in frame 208. (d) Target in frame 254. (1e) Target 1 in frame 445. (2e) Target 2 in frame 445. (f) Target in frame 815.

second-stage neural network structure model first uses a feature extractor to generate a series of preset anchors, which may contain objects to be detected, and then applies filtering rules to generate detection boxes. Compared with the Yolov5, the number of correct detection performs better with higher false alarms due to the relatively simple information provided by the SAR images of the CSL Video-SAR-GMTI mode.

MSCA refers to adding an attention mechanism only in the middle layer without shallow feature extraction; it performs better in video SAR moving targets detection by adding spatial and channelwise attention joint attention block in the middle layer. When the image is marked, the network focuses only on a small part of the image. Spatial attention can realize the allocation of more attention to the large weighted part so that the model can concentrate more on the interested content. After weight recalibration of each pixel in the feature map, channelwise attention transfers attention to the channels. In moving targets detection, channelwise attention will aggrandize the channel weights generated by the convolutional kernel extracted from target features. Channelwise attention can effectively extract semantic information from images. The number of correctly detected targets rises to 502, and the false alarm decreases to 165, which is benefited from the MSCA block. MSCA-SFCM adds shallow layer features extraction on the basis of MSCA. Shallow layers have reliable advantages in extracting location and edge information. Recalibrate the weight of shallow feature maps, and concat the results with deeper features, which can effectively achieve the fusion effect of shallow and deep features. In addition, the partial cross-stage operation is introduced in the middle layers to effectively improve the capability of the network. Accordingly, the number of missed targets drops to 93, and the false alarm targets are effectively reduced to 95.

In order to further exhibit the differences of target detection results, 6210 video SAR images are fed into the aforementioned five networks, while the partial results are presented in Fig. 13. The serial number $\{ai, bi, ci, di, ei, fi\}$ represent the six different images that are input to the neural network detector, and $i = \{1, 2, 3, 4, 5\}$ corresponds to HOG + SVM, Faster-RCNN, Yolov5, MSCA, and MSCA-SFCM detectors, respectively.

According to the application of parameter ER, the intensity comparison between the targets and the background clutter can be well simulated in the actual situation. Therefore, the discontinued bright lines of targets arising from the influence of background clutter can also be detected, as shown in *ci* and *ei* in Fig. 13. Since the input video SAR image is a gray-scale image, the reflectivity of the target in the background scene varies, which can be concretely manifested by the bright areas and dark areas in the image. For targets in the bright region, the features are relatively less obvious than those in the dark region, but they can still be detected accurately. Some buildings may show the same scattering line shape and, therefore, be mistakenly detected as targets, as shown in Fig. 13(f1)–(f5), which is the main cause for false alarm detection. Fig. 13(e1)–(e5) illustrates the result of multiple targets in one image; the deep learning network also shows excellent detection performance.

Compared to the Yolov5 moving targets detection in small targets, the MSCA and MSCA-SFCM introduce the attention module and the shallow feature of the low layer for getting better detection results. The targets in Fig. 13(a3) and (d3) are missed by Yolov5 but can be correctly detected by MSCA and MSCA-SFCM, as shown in Fig. 13(a4) and (d4) and (a5) and (d5), respectively. Faster-RCNN also has a better performance in detecting small targets, as shown in Fig. 13(a2) and (d2). Faster-RCNN also has a better performance in detecting small targets, as shown in Fig. 13(a2) and (d2). However, due to the relatively simple information provided by SAR images of the CSL Video-SAR-GMTI mode, false alarms of Faster-RCNN are relatively higher, while MSCA-SFCM shows good

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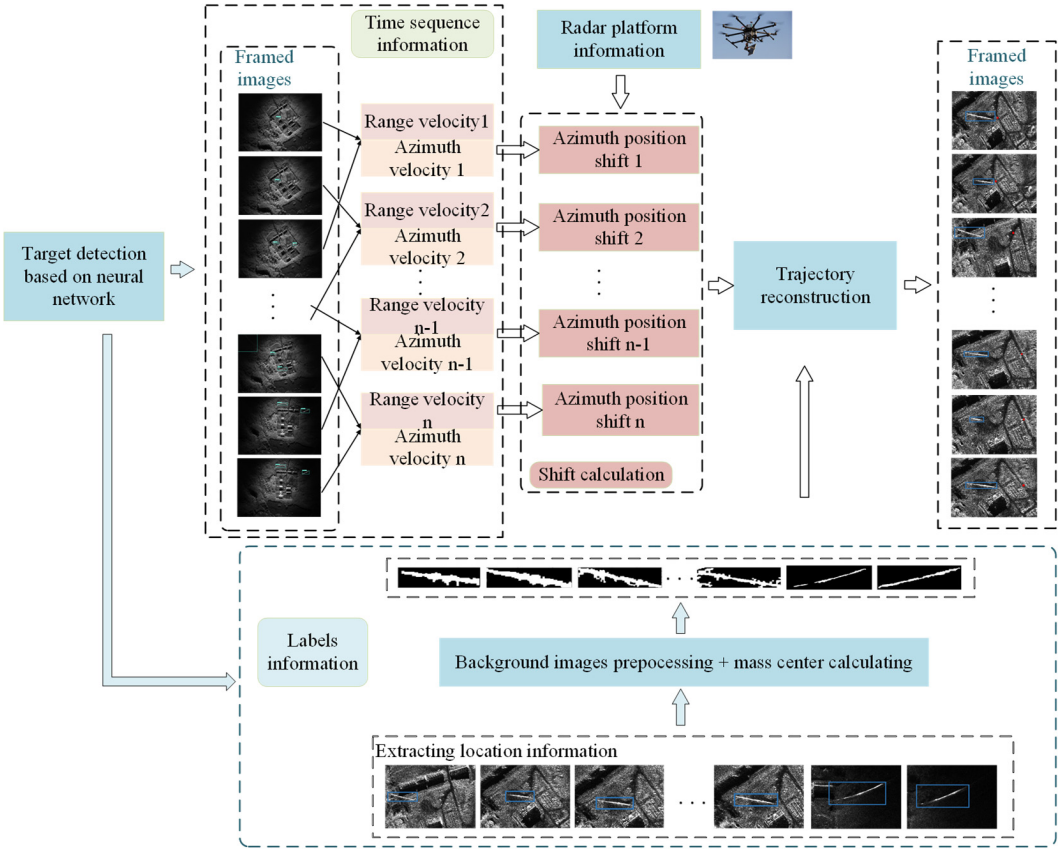


Fig. 15. Algorithm of target relocation.

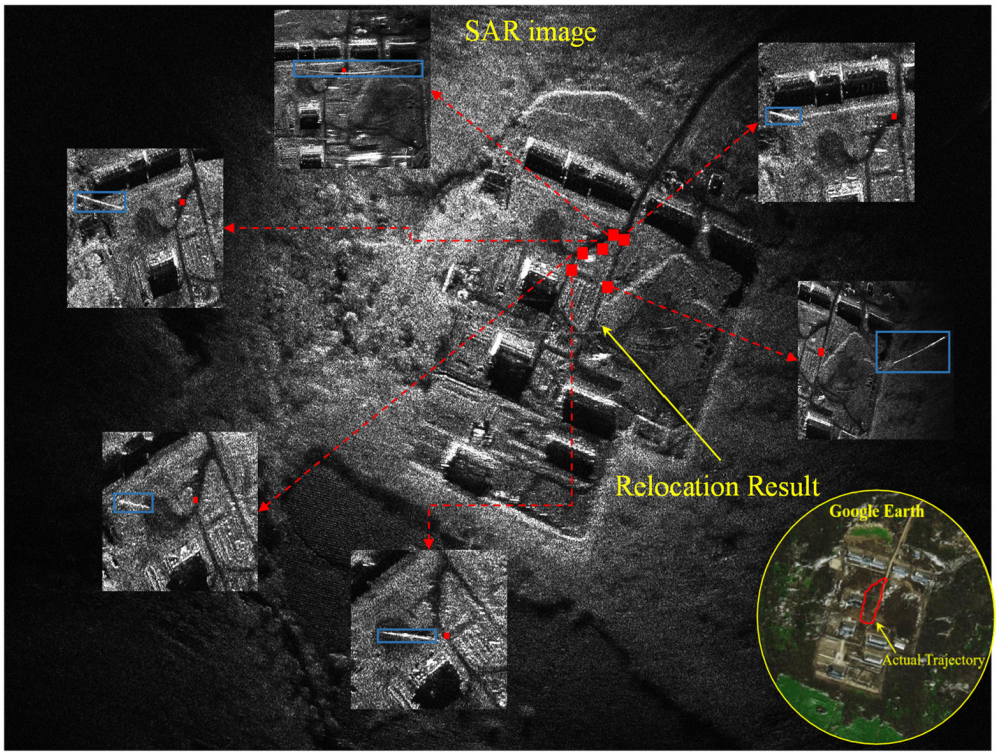


Fig. 16. Partial results of moving target relocation based on time sequence information of the detected bright line feature.

performance in reducing the false alarms due to the fusion of shallow feature and deep semantic information, as shown in Fig. 13(a2) and (a5), (c2) and (c5), and (e2) and (e5), respectively.

To further evaluate the effects of the improved network, the confidence of the targets with different networks is compared in Fig. 14. Each line in Fig. 14 represents the confidence changes of the seven targets that are detected by the same

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network structure corresponding to Fig. 13. In image (e), there are two targets, denoted as (1e) and (2e), respectively. The false alarms are represented by every single point, and different networks are shown in different colors.

Note that the MSCA network has significantly improved the rate of correctly moving targets detection, and the confidence of each target has been increased to a certain extent in all images. For example, the confidence of the target in image (b) increases from 0.56 to 0.73, and that of the target in (2e) also increases from 0.5 to 0.72. The final detection result of moving targets detection in the real SAR video is displayed on the gray line. It is obvious that the average level of confidence is greatly improved. In addition, it can be learned from false alarms that the confidence values are very low, most of which are even far below the detection confidence of real targets. False alarms can be reduced by increasing the confidence threshold of network detection. Nevertheless, if the confidence threshold is set too high, the number of correct detection results of targets will drop. For example, in the MSCA-SFCM, the detection result of image (d) is 0.44, while the false detection probability of image (c) can reach 0.59. When the confidence threshold is set at 0.6, the target in image (d) will not be detected. Accordingly, an appropriate increase in the confidence threshold during detection contributes to the reduced false alarm targets to some extent.

After moving target detection, the change of the position of the detected bright line between adjacent frames can be used to realize moving target relocation (it is the important information provided by the CSL Video-SAR-GMTI mode, which is unattainable for the traditional SAR-GMTI mode), which is the important research direction for our current work. In the following, we give some initial moving target relocation results to show the moving target relocation ability of the CSL Video-SAR-GMTI mode. The algorithm for target relocation is presented in Fig. 15.

In addition, 155 frames of the measured SAR data are tested for moving target relocation, and partial relocation results are shown in Fig. 16. Compared to the actual trajectory of the cooperative moving target, the moving target relocation algorithm works well in real data processing of the CSL Video-SAR-GMTI mode.

VII. Conclusion

When applying the deep learning method for moving targets detection in a real radar system, one of the main problems is that there are not enough moving targets samples for the network training, especially for the moving targets on the ground. With a small sample for neural network training, the detection performance and generalization ability will be greatly degraded. Therefore, in this article, we try to construct the training datasets with high-fidelity simulation and test the training results with real radar data obtained by our NUAA MiniSAR system. The proposed approach can be extended to other radar systems to alleviate the small data problem in deep learning-based target detection. Besides, we propose a novel MSCA-SFCM for moving target detection in the CSL Video-SAR-GMTI mode of our NUAA MiniSAR system. The processing results show that, combing with the simulated training data set, the proposed method has a good moving target detection performance compared with HOG + SVM, Faster-RCNN, and Yolov5.

However, limited by this article space, this article only focuses on the training data construction and moving target detection in a single frame of the video SAR image. In the future, we will pay more attention to the effective use of time sequences information of the video SAR system to further reduce the number of false alarms and missing targets, and study the target relocation method in depth based on the bright line feature of moving targets.

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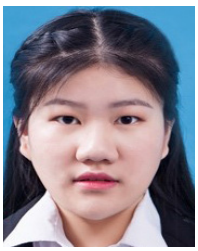
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